Trust infrastructure for cross-border B2B escrow.

TradeLock turns escrow from a static demo into a live operation: managed wallets, Arbitrum Sepolia contracts, proof uploads, dispute mapping, and a judge-ready dashboard.

- Arbitrum Sepolia
- Escrow + tUSD
- QStash automation
- Supabase + Redis
- Pinata/IPFS proofs

Live app: tradelock-pi.vercel.app

THE PROBLEM

International trade still fails at the moment trust matters most.

Buyers fear paying too early

They need proof that goods, inspection, or shipment milestones are real before funds move.

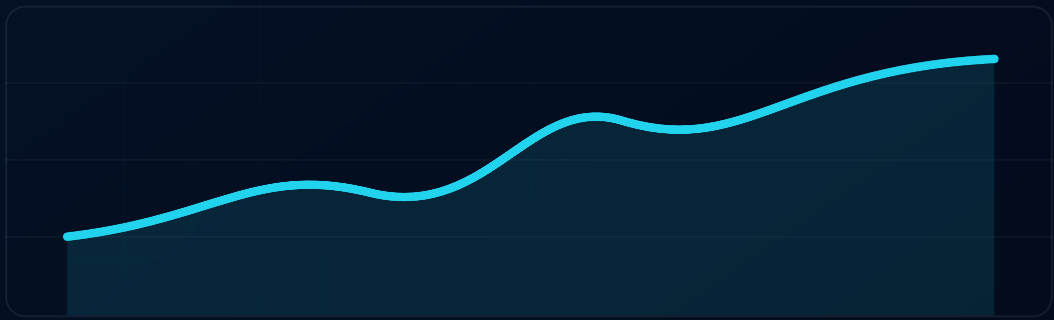
Sellers fear shipping too early

They need confidence that payment is locked and settlement cannot be quietly walked back.

Arbitrators lack evidence context

When disputes happen, chats and screenshots are not enough for a clean review path.

The gap is not only escrow. The gap is operational visibility: who funded, who uploaded proof, what evidence exists, where funds are frozen, and which on-chain event proves it.



THE SOLUTION

A live escrow workspace with proof, custody, and arbitration built in.

1

Create deal

Buyer and seller are mapped to known company profiles and wallets.

2

Fund escrow

tUSD is approved and locked through the escrow contract path.

3

Upload proof

Shipment, inspection, or milestone evidence is pinned to IPFS.

4

Release or dispute

Funds settle to seller or remain frozen for structured review.

Managed wallets

20

buyer, seller, arbitrator pool

Core asset

tUSD

test settlement token

Automation

24/7

QStash scheduled activity

Evidence

IPFS

Pinata proof storage

ARCHITECTURE

The system runs as a connected custody, state, proof, and contract loop.

A

QStash

Schedules activity and daily user growth.

B

Custodial Engine

Funds and orchestrates managed wallet flows.

C

Redis

Fast custody registry and live cache.

D

Supabase

Deals, disputes, audit events, settings.

E

Pinata/IPFS

Proof file persistence and review links.

F

Contracts

Escrow and tUSD settlement on Arbitrum Sepolia.

The dashboard reads from the same state path used by automation, so reviewers see changing custody activity, updated deals, and dispute-ready evidence instead of mock-only rows.

JUDGE MODE

A guided review surface so judges can verify claims fast.

Demo checklist

Dashboard, Deals, Disputes, Audit Trail, Counterparties, and Settings are mapped into one review path.

Counterparty mapping

Companies, roles, countries, wallets, trust scores, and deal counts are visible in one table.

Evidence trail

Latest deal, proof hash, transaction hash, open disputes, and service health are surfaced immediately.

/?screen=judge

HACKATHON REVIEW COCKPIT

Everything judges need in one place.

Live state, counterparties, disputes, proof trail, architecture, and deck links.

Managed wallets

20

Open disputes

Live

DISPUTE AND PROOF MODEL

Arbitration is mapped before conflict becomes chaos.

Deal-linked disputes

Every dispute references a deal ID, buyer, seller, amount, evidence status, update time, and transaction hash.

Frozen value visibility

Open disputes show funds that remain frozen in escrow until review is resolved.

Proof files

IPFS-backed evidence such as photos, inspection notes, manifests, and completion certificates can be attached to audit events.

Audit trail

Deal created, deposit funded, proof uploaded, proof verified, released, and dispute events stay searchable and exportable.

1

Issue raised

Buyer or seller opens dispute.

2

Funds frozen

Escrow state prevents premature release.

3

Evidence reviewed

Files, proof hash, and tx context are inspected.

4

Resolution path

Release, escalation, or archive can be reasoned about cleanly.

AUTOMATION LAYER

The app keeps moving even when no one clicks the demo.

Recurring market activity

QStash calls /api/cron/activity to simulate active B2B escrow flow.

Daily user growth

/api/cron/daily-user adds managed wallet activity to make the network feel alive.

Live dashboard sync

The UI refreshes app state, custody snapshot, and system health on a tight cadence.

Status

Live

automation-ready app

Cache

Redis

managed wallet state

Store

SQL

Supabase app state

Network

ARB

Sepolia testnet

WHY IT WINS

TradeLock looks and behaves like real escrow operations.

The product is strong because the UI, automation, and backend all tell the same story: escrow is not just a contract call. It is a workflow with counterparties, proof, evidence, auditability, and trust recovery.

Operational credibility

Live custody state and recurring activity make the demo feel used, not staged.

Judge-friendly proof

Judge Mode connects every claim to a screen and evidence source.

Portable architecture

The same pattern can move from testnet tUSD to production settlement assets later.

TradeLock dashboard

SELECTED DEAL

Buyer -> Escrow -> Seller

Milestone proof, IPFS hash, transaction hash, status, and dispute path are all visible.

ROADMAP

From testnet escrow cockpit to production trade rails.

Phase 1: Hackathon proof

Live Arbitrum Sepolia app, Judge Mode, proof uploads, disputes, automation, and managed wallet activity.

Phase 2: Production pilots

Role-based access, richer evidence templates, importer/exporter onboarding, and contract hardening.

Phase 3: Trade network

Multi-asset settlement, reputation graph, arbitration marketplace, and compliance-grade audit export.

1

Lock

Buyer funds escrow.

2

Prove

Seller submits evidence.

3

Review

Judge or arbitrator inspects the trail.

4

Settle

Funds release or remain protected.

TradeLock is the credible middle layer for global B2B trust: simple enough to demo, structured enough to become real infrastructure.